



# Chapter 10: Big Data

**Database System Concepts, 7<sup>th</sup> Ed.**

©Silberschatz, Korth and Sudarshan

See [www.db-book.com](http://www.db-book.com) for conditions on re-use



# Motivation

- Very large volumes of data being collected
  - Driven by growth of web, social media, and more recently internet-of-things
  - Web logs were an early source of data
    - Analytics on web logs has great value for advertisements, web site structuring, what posts to show to a user, etc
- Big Data: differentiated from data handled by earlier generation databases
  - **Volume**: much larger amounts of data stored
  - **Velocity**: much higher rates of insertions
  - **Variety**: many types of data, beyond relational data



# Querying Big Data

- Transaction processing systems that need very high scalability
  - Many applications willing to sacrifice ACID properties and other database features, if they can get very high scalability
- Query processing systems that
  - Need very high scalability, and
  - Need to support non-relation data



# Big Data Storage Systems

- Distributed file systems
- Sharding across multiple databases
- Key-value storage systems
- Parallel and distributed databases



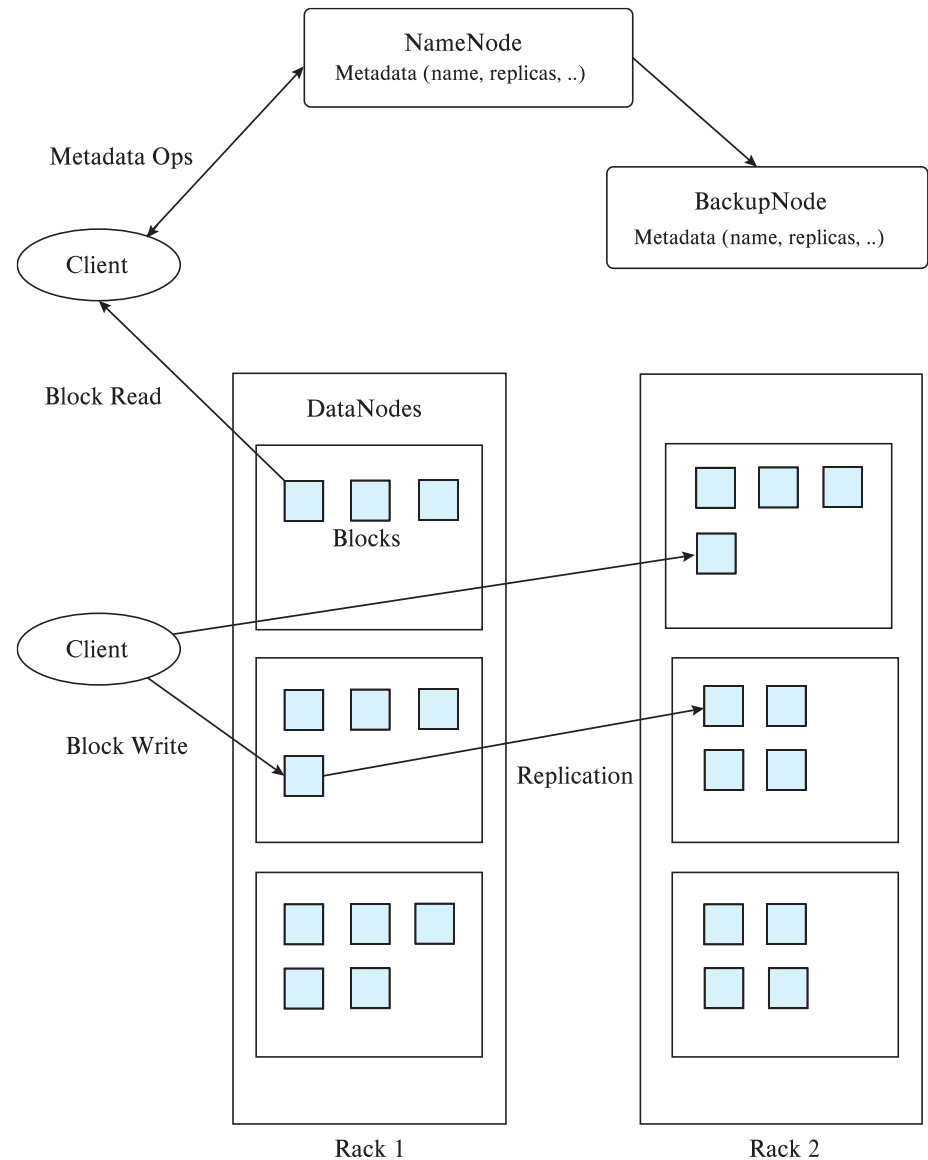
# Distributed File Systems

- A distributed file system stores data across a large collection of machines, but provides single file-system view
- Highly scalable distributed file system for large data-intensive applications.
  - E.g., 10K nodes, 100 million files, 10 PB
- Provides redundant storage of massive amounts of data on cheap and unreliable computers
  - Files are replicated to handle hardware failure
  - Detect failures and recovers from them
- Examples:
  - Google File System (GFS)
  - Hadoop File System (HDFS)



# Hadoop File System Architecture

- Single Namespace for entire cluster
- Files are broken up into blocks
  - Typically 64 MB block size
  - Each block replicated on multiple DataNodes
- Client
  - Finds location of blocks from NameNode
  - Accesses data directly from DataNode





# Hadoop Distributed File System (HDFS)

- **NameNode**
  - Maps a filename to list of Block IDs
  - Maps each Block ID to DataNodes containing a replica of the block
- **DataNode**: Maps a Block ID to a physical location on disk
- Data Coherency
  - Write-once-read-many access model
  - Client can only append to existing files
- Distributed file systems good for millions of large files
  - But have very high overheads and poor performance with billions of smaller tuples



# Sharding

- **Sharding**: partition data across multiple databases
- Partitioning usually done on some ***partitioning attributes*** (also known as ***partitioning keys*** or ***shard keys*** e.g. user ID
  - E.g., records with key values from 1 to 100,000 on database 1, records with key values from 100,001 to 200,000 on database 2, etc.
- Application must track which records are on which database and send queries/updates to that database
- Positives: scales well, easy to implement
- Drawbacks:
  - Not transparent: application has to deal with routing of queries, queries that span multiple databases
  - When a database is overloaded, moving part of its load out is not easy
  - Chance of failure more with more databases
    - need to keep replicas to ensure availability, which is more work for application





# Parallel Databases and Data Stores

- Supporting scalable data access
  - Approach 1: memcache or other caching mechanisms at application servers, to reduce database access
    - Limited in scalability
  - Approach 2: Partition (“shard”) data across multiple separate database servers
  - Approach 3: Use existing parallel databases
    - Historically: parallel databases that can scale to large number of machines were designed for decision support not OLTP
  - Approach 4: Massively Parallel Key-Value Data Store
    - Partitioning, high availability etc. completely transparent to application
- Sharding systems and key-value stores don’t support many relational features, such as joins, integrity constraints, etc., across partitions.



# Key Value Storage Systems

- Key-value storage systems store large numbers (billions or even more) of small (KB-MB) sized records
- Records are **partitioned** across multiple machines and
- Queries are routed by the system to appropriate machine
- Records are also **replicated** across multiple machines, to ensure availability even if a machine fails
  - Key-value stores ensure that updates are applied to all replicas, to ensure that their values are **consistent**



# Key Value Storage Systems

- Key-value stores may store
  - **uninterpreted bytes**, with an associated key
    - E.g., Amazon S3, Amazon Dynamo
  - **Wide-table** (can have arbitrarily many attribute names) with associated key
    - Google BigTable, Apache Cassandra, Apache Hbase, Amazon DynamoDB
    - Allows some operations (e.g., filtering) to execute on storage node
  - JSON
    - MongoDB, CouchDB (document model)
- **Document stores** store semi-structured data, typically JSON
- Some key-value stores support multiple versions of data, with timestamps/version numbers



# Data Representation

- An example of a JSON object is:

```
{  
  "ID": "22222",  
  "name": {  
    "firstname": "Albert",  
    "lastname": "Einstein"  
  },  
  "deptname": "Physics",  
  "children": [  
    { "firstname": "Hans", "lastname": "Einstein" },  
    { "firstname": "Eduard", "lastname": "Einstein" }  
  ]  
}
```



# Key Value Storage Systems

- Key-value stores support
  - **put**(key, value): used to store values with an associated key,
  - **get**(key): which retrieves the stored value associated with the specified key
  - **delete**(key) -- Remove the key and its associated value
- Some systems also support **range queries** on key values
- Document stores also support queries on non-key attributes
  - See book for MongoDB queries
- Key value stores are not full database systems
  - Have no/limited support for transactional updates
  - Applications must manage query processing on their own
- Not supporting above features makes it easier to build scalable data storage systems
  - Also called **NoSQL** systems



# Parallel and Distributed Databases

- Parallel databases run multiple machines (cluster)
  - Developed in 1980s, well before Big Data
- Parallel databases were designed for smaller scale (10s to 100s of machines)
  - Did not provide easy scalability
- **Replication** used to ensure data availability despite machine failure
  - But typically restart query in event of failure
    - Restarts may be frequent at very large scale
    - Map-reduce systems (coming up next) can continue query execution, working around failures



# Replication and Consistency

- **Availability** (system can run even if parts have failed) is essential for parallel/distributed databases
  - Via replication, so even if a node has failed, another copy is available
- **Consistency** is important for replicated data
  - All live replicas have same value, and each read sees latest version
  - Often implemented using majority protocols
    - E.g., have 3 replicas, reads/writes must access 2 replicas
      - Details in chapter 23
- **Network partitions** (network can break into two or more parts, each with active systems that can't talk to other parts)
- In presence of partitions, cannot guarantee both availability and consistency
  - Brewer's CAP "Theorem"



# Replication and Consistency

- Very large systems will partition at some point
  - Choose one of consistency or availability
- Traditional database choose consistency
- Most Web applications choose availability
  - Except for specific parts such as order processing
- More details later, in Chapter 23





# The MapReduce Paradigm

- Platform for reliable, scalable parallel computing
- Abstracts issues of distributed and parallel environment from programmer
  - Programmer provides core logic (via `map()` and `reduce()` functions)
  - System takes care of parallelization of computation, coordination, etc.
- Paradigm dates back many decades
  - But very large scale implementations running on clusters with  $10^3$  to  $10^4$  machines are more recent
  - Google Map Reduce, Hadoop, ..
- Data storage/access typically done using distributed file systems or key-value stores



# MapReduce: Word Count Example

- Consider the problem of counting the number of occurrences of each word in a large collection of documents
- How would you do it in parallel?
- Solution:
  - Divide documents among workers
  - Each worker parses document to find all words, map function outputs (word, count) pairs
  - Partition (word, count) pairs across workers based on word
  - For each word at a worker, reduce function locally add up counts
- Given input: “One a penny, two a penny, hot cross buns.”
  - Records output by the map() function would be
    - (“One”, 1), (“a”, 1), (“penny”, 1), (“two”, 1), (“a”, 1), (“penny”, 1), (“hot”, 1), (“cross”, 1), (“buns”, 1).
  - Records output by reduce function would be
    - (“One”, 1), (“a”, 2), (“penny”, 2), (“two”, 1), (“hot”, 1), (“cross”, 1), (“buns”, 1)



# Pseudo-code of Word Count

**map**(String record):

    for each word in record  
        emit(word, 1);

// First attribute of emit above is called **reduce key**

// In effect, group by is performed on reduce key to create a

// list of values (all 1's in above code). This requires **shuffle step**

// across machines.

// The reduce function is called on list of values in each group

**reduce**(String key, List value\_list):

    String word = key

    int count = 0;

    for each value in value\_list:

        count = count + value

    Output(word, count);



# MapReduce Programming Model

- Inspired from map and reduce operations commonly used in functional programming languages like Lisp.
- Input: a set of key/value pairs
- User supplies two functions:
  - **map**(k,v)  $\rightarrow$  list(k1,v1)
  - **reduce**(k1, list(v1))  $\rightarrow$  v2
- (k1,v1) is an intermediate key/value pair
- Output is the set of (k1,v2) pairs
- For our example, assume that system
  - Breaks up files into lines, and
  - Calls map function with value of each line
    - Key is the line number



# MapReduce Example 2: Log Processing

- Given log file in following format:

...

2013/02/21 10:31:22.00EST [/slide-dir/11.ppt](#)

2013/02/21 10:43:12.00EST [/slide-dir/12.ppt](#)

2013/02/22 18:26:45.00EST [/slide-dir/13.ppt](#)

2013/02/22 20:53:29.00EST [/slide-dir/12.ppt](#)

...

- Goal: find how many times each of the files in the slide-dir directory was accessed between 2013/01/01 and 2013/01/31.
- Options:
  - Sequential program too slow on massive datasets
  - Load into database expensive, direct operation on log files cheaper
  - Custom built parallel program for this task possible, but very laborious
  - Map-reduce paradigm



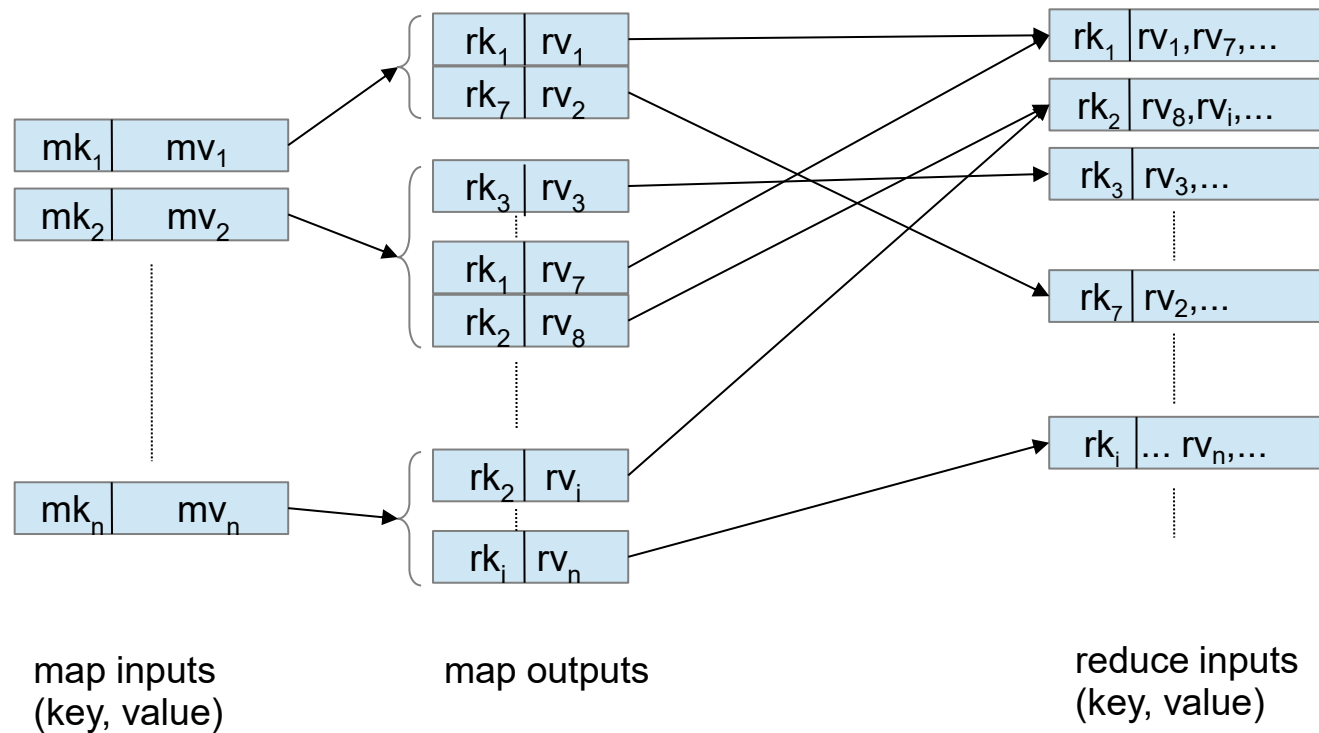
# MapReduce: File Access Count Example

```
map(String key, String record) {  
    String attribute[3];  
    .... break up record into tokens (based on space character), and store the  
        tokens in array attributes  
    String date = attribute[0];  
    String time = attribute[1];  
    String filename = attribute[2];  
    if (date between 2013/01/01 and 2013/01/31  
        and filename starts with "/slide-dir/")  
        emit(filename, 1).  
}  
  
reduce(String key, List recordlist) {  
    String filename = key;  
    int count = 0;  
    For each record in recordlist  
        count = count + 1.  
    output(filename, count)  
}
```



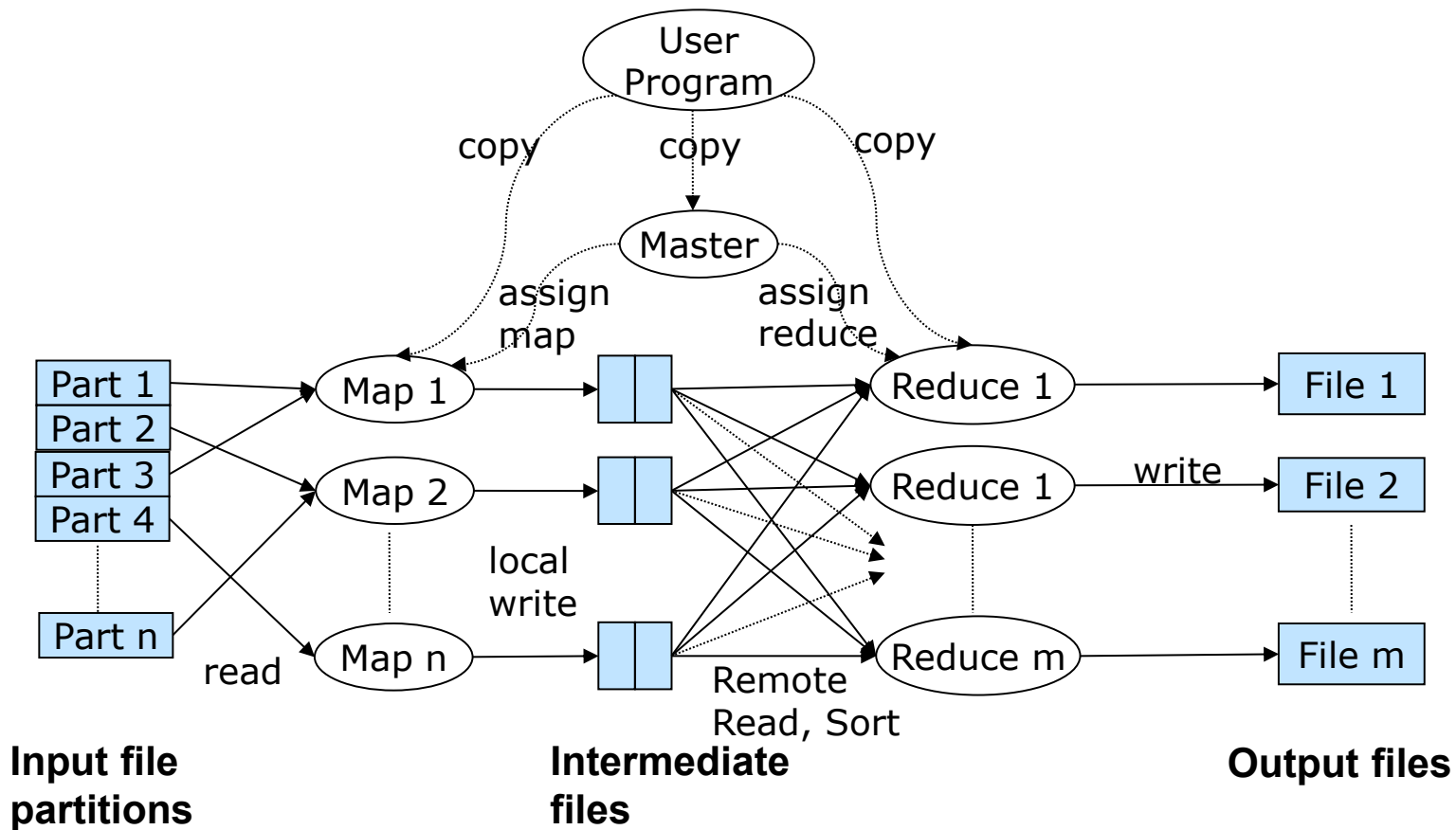
# Schematic Flow of Keys and Values

- Flow of keys and values in a map reduce task





# Parallel Processing of MapReduce Job







# Hadoop MapReduce

- Google pioneered map-reduce implementations that could run on thousands of machines (nodes), and transparently handle failures of machines
- Hadoop is a widely used open source implementation of Map Reduce written in Java
  - Map and reduce functions can be written in several different languages, we use Java.
- Input and output to map reduce systems such as Hadoop must be done in parallel
  - Google used GFS distributed file system
  - Hadoop uses Hadoop File System (HDFS),
  - Input files can be in several formats
    - Text/CSV
    - compressed representation such as Avro, ORC and Parquet
  - Hadoop also supports key-value stores such as Hbase, Cassandra, MongoDB, etc.



# Types in Hadoop

- Generic Mapper and Reducer interfaces both take four type arguments, that specify the types of the
  - input key, input value, output key and output value
- Map class in next slide implements the Mapper interface
  - Map input key is of type LongWritable, i.e. a long integer
  - Map input value which is (all or part of) a document, is of type Text.
  - Map output key is of type Text, since the key is a word,
  - Map output value is of type IntWritable, which is an integer value.



# Hadoop Code in Java: Map Function

```
public static class Map extends Mapper<LongWritable, Text, Text, IntWritable>
{
    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException
    {
        String line = value.toString();
        StringTokenizer tokenizer = new StringTokenizer(line);
        while (tokenizer.hasMoreTokens()) {
            word.set(tokenizer.nextToken());
            context.write(word, one);
        }
    }
}
```



# Hadoop Code in Java: Reduce Function

```
public static class Reduce extends Reducer<Text, IntWritable, Text, IntWritable> {  
    public void reduce(Text key, Iterable<IntWritable> values,  
        Context context) throws IOException, InterruptedException  
    {  
        int sum = 0;  
        for (IntWritable val : values) {  
            sum += val.get();  
        }  
        context.write(key, new IntWritable(sum));  
    }  
}
```



# Hadoop Job Parameters

- The classes that contain the map and reduce functions for the job
  - Set by methods `setMapperClass()` and `setReducerClass()`
- The types of the job's output key and values
  - Set by methods `setOutputKeyClass()` and `setOutputValueClass()`
- The input format of the job
  - Set by method `job.setInputFormatClass()`
    - Default input format in Hadoop is the `TextInputFormat`,
      - Map key whose value is a byte offset into the file, and
      - Map value is the contents of one line of the file
- The directories where the input files are stored, and where the output files must be created
  - Set by `addInputPath()` and `addOutputPath()`
- And many more parameters



# Hadoop Code in Java: Overall Program

```
public class WordCount {  
    public static void main(String[] args) throws Exception {  
        Configuration conf = new Configuration();  
        Job job = new Job(conf, "wordcount");  
        job.setOutputKeyClass(Text.class);  
        job.setOutputValueClass(IntWritable.class);  
        job.setMapperClass(Map.class);  
        job.setReducerClass(Reduce.class);  
        job.setInputFormatClass(TextInputFormat.class);  
        job.setOutputFormatClass(TextOutputFormat.class);  
        FileInputFormat.addInputPath(job, new Path(args[0]));  
        FileOutputFormat.setOutputPath(job, new Path(args[1]));  
        job.waitForCompletion(true);  
    }  
}
```



# Local Pre-Aggregation

- **Combiners**: perform partial aggregation to minimize network traffic
  - E.g., within machine
  - And/or at rack level
- In Hadoop, reduce function is used by default if combiners are enabled
  - But alternative implementation of combiner can be specified if input and output types of reducers are different



# Map Reduce vs. Databases

- Map Reduce widely used for parallel processing
  - Google, Yahoo, and 100' s of other companies
  - Example uses: compute PageRank, build keyword indices, do data analysis of web click logs, ....
  - Allows procedural code in map and reduce functions
  - Allows data of any type
- Many real-world uses of MapReduce cannot be expressed in SQL
- But many computations are much easier to express in SQL
  - Map Reduce is cumbersome for writing simple queries





# Map Reduce vs. Databases (Cont.)

- Relational operations (select, project, join, aggregation, etc.) can be expressed using Map Reduce
- SQL queries can be translated into Map Reduce infrastructure for execution
  - Apache Hive SQL, Apache Pig Latin, Microsoft SCOPE
- Current generation execution engines support not only Map Reduce, but also other algebraic operations such as joins, aggregation, etc. natively.



# BEYOND MAPREDUCE: ALGEBRAIC OPERATIONS



# Algebraic Operations

- Current generation execution engines
  - natively support algebraic operations such as joins, aggregation, etc. natively.
  - Allow users to create their own algebraic operators
  - Support trees of algebraic operators that can be executed on multiple nodes in parallel
- E.g. Apache Tez, Spark
  - Tez provides low level API; Hive on Tez compiles SQL to Tez
  - Spark provides more user-friendly API



# Algebraic Operations in Spark

- **Resilient Distributed Dataset (RDD)** abstraction
  - Collection of records that can be stored across multiple machines
- RDDs can be created by applying algebraic operations on other RDDs
- RDDs can be lazily computed when needed
- Spark programs can be written in Java/Scala/R
  - Our examples are in Java
- Spark makes use of Java 8 Lambda expressions; the code  
`s -> Arrays.asList(s.split(" ")).iterator()`  
defines unnamed function that takes argument `s` and executes the expression `Arrays.asList(s.split(" ")).iterator()` on the argument
- Lambda functions are particularly convenient as arguments to map, reduce and other functions



# Word Count in Spark

```
SparkSession spark =  
    SparkSession.builder().appName("WordCount").getOrCreate();  
  
JavaRDD<String> lines = spark.read().textFile(args[0]).javaRDD();  
JavaRDD<String> words = lines.flatMap(s -> Arrays.asList(s.split(" ")).iterator());  
JavaPairRDD<String, Integer> ones = words.mapToPair(s -> new Tuple2<>(s, 1));  
JavaPairRDD<String, Integer> counts = ones.reduceByKey((i1, i2) -> i1 + i2);  
  
counts.saveAsTextFile("outputDir"); // Save output files in this directory  
  
List<Tuple2<String, Integer>> output = counts.collect();  
for (Tuple2<String,Integer> tuple : output) {  
    System.out.println(tuple);  
}  
spark.stop();
```



# Algebraic Operations in Spark

- Algebraic operations in Spark are typically executed in parallel on multiple machines
  - With data partitioned across the machines
- Algebraic operations are executed lazily, not immediately
  - Our preceding program creates an operator tree
  - Tree is executed only on specific functions such as `saveAsTextFile()` or `collect()`
  - Query optimization can be performed on tree before it is executed



# Spark DataFrames and DataSet

- RDDs in Spark can be typed in programs, but not dynamically
- The DataSet type allows types to be specified dynamically
- Row is a row type, with attribute names
  - In code below, attribute names/types of instructor and department are inferred from files read
- Operations filter, join, groupBy, agg, etc defined on DataSet, and can execute in parallel
- ```
Dataset<Row> instructor = spark.read().parquet("...");
Dataset<Row> department = spark.read().parquet("...");
instructor.filter(instructor.col("salary").gt(100000))
.join(department, instructor.col("dept name")
.equalTo(department.col("dept name")))
.groupBy(department.col("building"))
.agg(count(instructor.col("ID")));
```



# STREAMING DATA





# Streaming Data and Applications

- **Streaming data** refers to data that arrives in a continuous fashion
  - Contrast to **data-at-rest**
- Applications include:
  - Stock market: stream of trades
  - e-commerce site: purchases, searches
  - Sensors: sensor readings
    - Internet of things
  - Network monitoring data
  - Social media: tweets and posts can be viewed as a stream
- Queries on streams can be very useful
  - Monitoring, alerts, automated triggering of actions



# Querying Streaming Data

Approaches to querying streams:

- **Windowing**: Break up stream into windows, and queries are run on windows
  - Stream query languages support window operations
  - Windows may be based on time or tuples
  - Must figure out when all tuples in a window have been seen
    - Easy if stream totally ordered by timestamp
    - **Punctuations** specify that all future tuples have timestamp greater than some value
- **Continuous Queries**: Queries written e.g. in SQL, output partial results based on stream seen so far; query results updated continuously
  - Have some applications, but can lead to flood of updates



# Querying Streaming Data (Cont.)

Approaches to querying streams (Cont.):

- **Algebraic operators on streams:**
  - Each operator consumes tuples from a stream and outputs tuples
  - Operators can be written e.g., in an imperative language
  - Operator may maintain state
- **Pattern matching:**
  - Queries specify patterns, system detects occurrences of patterns and triggers actions
  - **Complex Event Processing (CEP)** systems
  - E.g., Microsoft StreamInsight, Flink CEP, Oracle Event Processing



# Stream Processing Architectures

- Many stream processing systems are purely in-memory, and do not persist data
- **Lambda architecture**: split stream into two, one output goes to stream processing system and the other to a database for storage
  - Easy to implement and widely used
  - But often leads to duplication of querying effort, once on streaming system and once in database



# Stream Extensions to SQL

- SQL Window functions described in Section 5.5.2
- Streaming systems often support more window types
  - **Tumbling window**
    - E.g., hourly windows, windows don't overlap
  - **Hopping window**
    - E.g., hourly window computed every 20 minutes
  - **Sliding window**
    - Window of specified size (based on timestamp interval or number of tuples) around each incoming tuple
  - **Session window**
    - Groups tuples based on user sessions



# Window Syntax in SQL

- Windowing syntax varies widely by system
- E.g., in Azure Stream Analytics SQL:

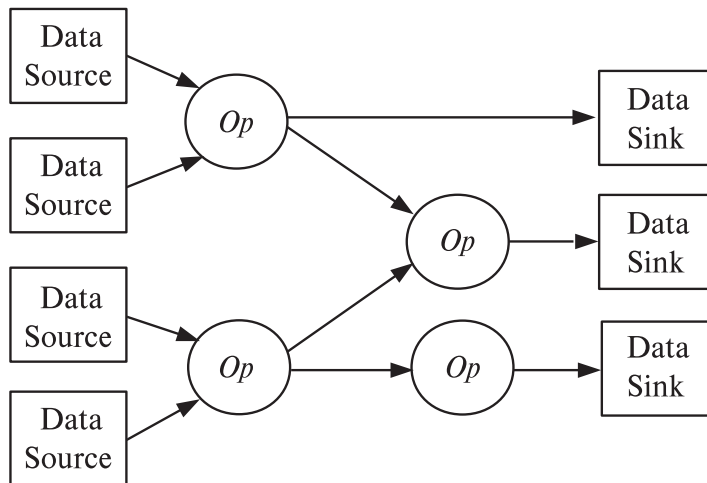
```
select item, System.Timestamp as window end, sum(amount)  
from order timestamp by datetime  
group by itemid, tumblingwindow(hour, 1)
```

- Aggregates are applied on windows
- Result of windowing operation on a stream is a relation
- Many systems support stream-relation joins
- Stream-stream joins often require join conditions to specify bound on timestamp gap between matching tuples
  - E.g., tuples must be at most 30 minutes apart in timestamp

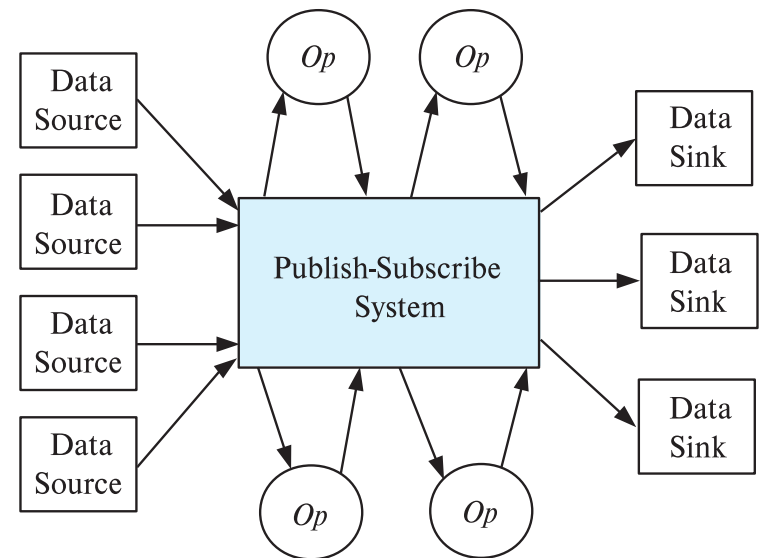


# Algebraic Operations on Streams

- Tuples in streams need to be routed to operators
- Routing of streams using DAG and publish-subscribe representations
  - Used in Apache Storm and Apache Kafka respective



(a) DAG representation of streaming data flow



(b) Publish-subscribe representation of streaming data flow



# Publish Subscribe Systems

- **Publish-subscribe (pub-sub)** systems provide convenient abstraction for processing streams
  - Tuples in a stream are published to a topic
  - Consumers subscribe to topic
- Parallel pub-sub systems allow tuples in a topic to be partitioned across multiple machines
- **Apache Kafka** is a popular parallel pub-sub system widely used to manage streaming data
- More details in book





# GRAPH DATABASES



# Graph Data Model

- Graphs are a very general data model
- ER model of an enterprise can be viewed as a graph
  - Every entity is a node
  - Every binary relationship is an edge
  - Ternary and higher degree relationships can be modelled as binary relationships



# Graph Data Model (Cont.)

- Graphs can be modelled as relations
  - *node(ID, label, node\_data)*
  - *edge(fromID, toID, label, edge\_data)*
- Above representation too simplistic
- Graph databases like Neo4J can provide a **graph view of relational schema**
  - Relations can be identified as representing either nodes or edges
- Query languages for graph databases make it
  - easy to express queries requiring edge traversal
  - allow efficient algorithms to be used for evaluation



# Graph Data Model (Cont.)

- Suppose
  - Relations *instructor* and *student* are nodes, and
  - Relation *advisor* represents edges between instructors and student
- Query in Neo4J:  
**match** (*i:instructor*)<-[:*advisor*]-(*s:student*)  
**where** *i.dept name*= 'Comp. Sci.'  
**return** *i.ID* **as** ID, *i.name* **as** name, **collect**(*s.name*) **as** advisees
- **match** clause matches nodes and edges in graphs
- Recursive traversal of edges is also possible
  - Suppose *prereq(course\_id, prereq\_id)* is modeled as an edge
  - Transitive closure can be done as follows:

```
match (c1:course)-[:prereq *1..]->(c2:course)  
return c1.course id, c2.course id
```



# Parallel Graph Processing

- Very large graphs (billions of nodes, trillions of edges)
  - Web graph: web pages are nodes, hyper links are edges
  - Social network graph: people are nodes, friend/follow links are edges
- Two popular approaches for parallel processing on such graphs
  - Map-reduce and algebraic frameworks
  - **Bulk synchronous processing (BSP)** framework
- Multiple iterations are required for any computations on graphs
  - Map-reduce/algebraic frameworks often have high overheads per iteration
  - BSP frameworks have much lower per-iteration overheads
- Google's Pregel system popularized the BSP framework
- Apache Giraph is an open-source version of Pregel
- Apache Spark's GraphX component provides a Pregel-like API



# Bulk Synchronous Processing

- Each vertex (node) of a graph has data (state) associated with it
  - Vertices are partitioned across multiple machines, and state of node kept in-memory
- Analogous to map() and reduce() functions, programmers provide methods to be executed for each node
  - Node method can send messages to or receive messages from neighboring nodes
- Computation consists of multiple iterations, or supersteps
- In each **superstep**
  - Nodes process received messages
  - Update their state, and
  - Send further messages or vote to halt
  - Computation ends when all nodes vote to halt, and there are no pending messages;



# End of Chapter 10

